# Radial Basis Function (RBF) Kernel

$[e^{-\gamma \|a-b\|^2}]$

*Example-3*

$\begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$

**Normal Model**

Calculate $\mathbf{a^T b}$

$\begin{pmatrix} a_1 & a_2 & b_1 \\ & & b_2 \end{pmatrix}$

*Map Input data from 2-D to infinite Dimension*

$\begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$

$e^{-\gamma \|a-b\|^2}$

**Vectors Mapping**

$\mathbf{a, b \Rightarrow u, v}$

*Higher Computational Complexity !!!*

$\begin{matrix} u_1 & v_1 \\ u_2 & v_2 \\ u_3 & v_3 \\ u_{\dots} & v_{\dots} \\ u_{\dots} & v_{\dots} \\ u_{\infty} & v_{\infty} \end{matrix}$

**Kernelized Model**

Calculate $\mathbf{u^T v}$

$\begin{pmatrix} u_1 & u_2 & u_3 & \dots & \dots & u_{\infty} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_{\dots} \\ v_{\dots} \\ v_{\infty} \end{pmatrix}$

*Higher Computational Complexity !!!*